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Recommending Career Path for Masters' Graduates

Business Intelligence and Analytics

Project Proposal

Presented By: Group 2

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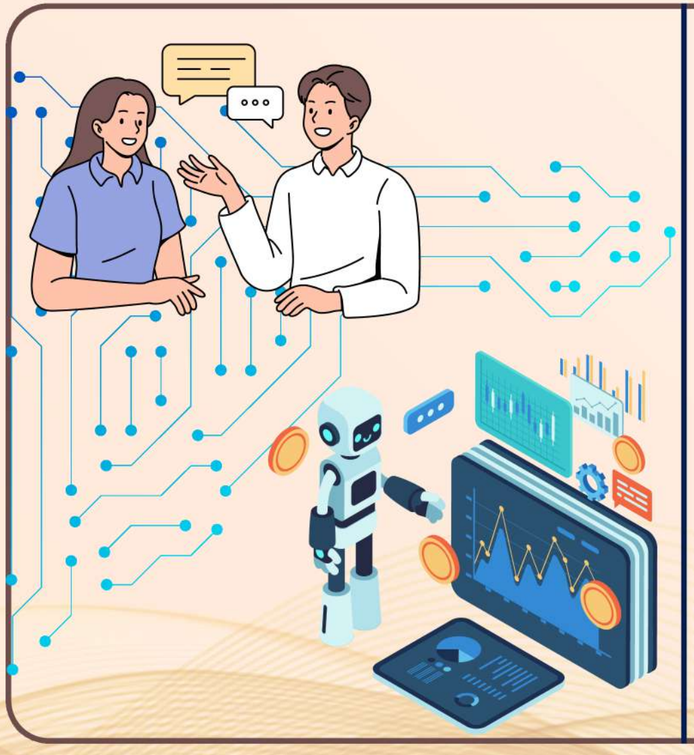
THE PROBLEM:
THE "STRATEGIC DECISION" GAP



1

THE SHIFT:

Choosing a Master's degree is now a strategic decision rather than just academic

2

THE UNCERTAINTY:

AI, automation, and shifting demands create job role instability

3

THE GAP:

Current guidance relies on personal stories or short-term salary signals rather than data

PRIMARY OBJECTIVES

- Identify and group Master's degree categories
- Map degrees to realistic, data-driven career paths
- Construct a Graduate Career Stability Index (GCSI)

CATEGORIZE → **MAP** → **EVALUATE**

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DATA SOURCES

24,084
Master's programs
(USA)

1394
Occupations
(Salary)

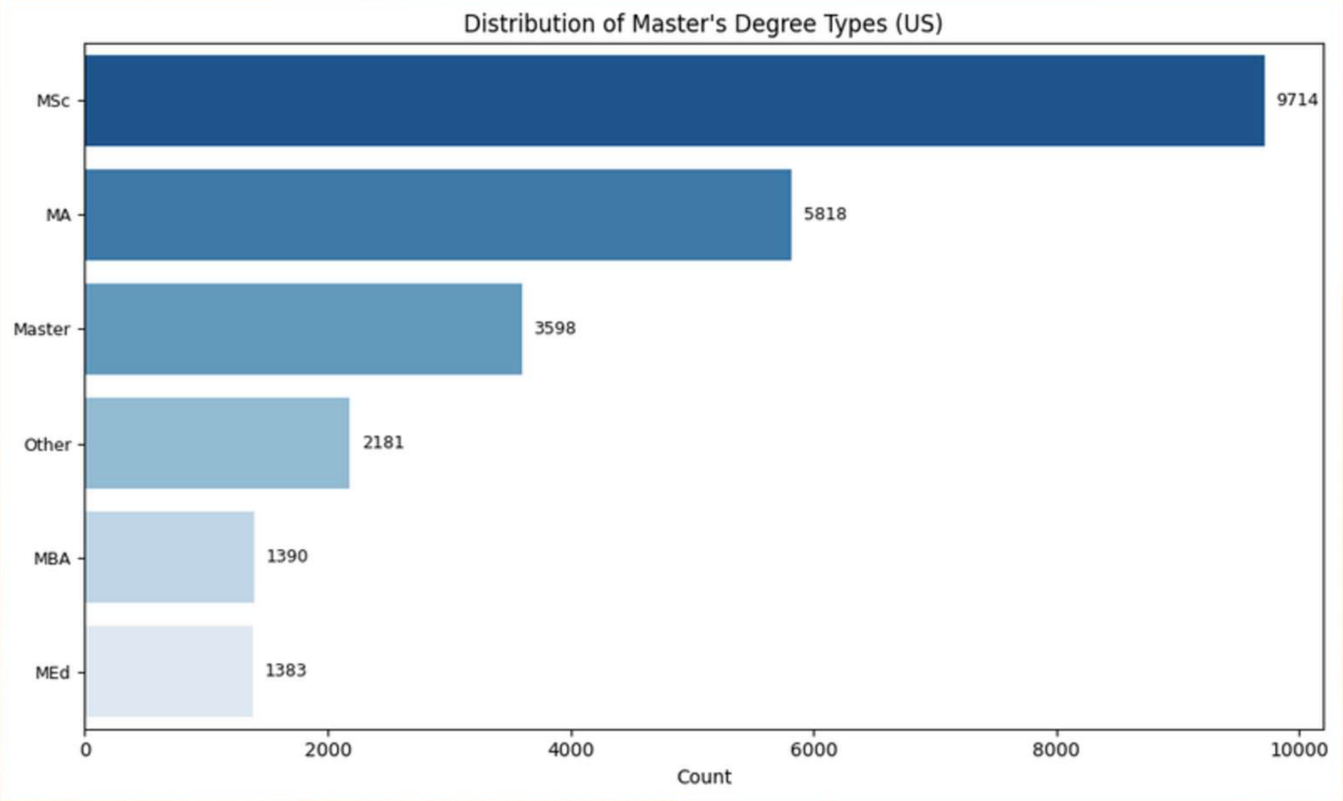
702
Occupations
(Automation)

Dataset	Source (Kaggle)	Role in project
1. Master's programs	Masters Portal Programs	Define degree categories and professional pathways (US scope)
2. Automation	Occupation, Likelihood of Automation	Measure long-term occupational vulnerability (automation probability)
3. Salary	Occupation, Salary	Capture economic value, employment scale, and wage dispersion

Exploratory Data Analysis (EDA)

Master's programs

Total US Programs: 24,084



Key Observations:

- **Master of Science (MSc)** dominates the US master's landscape
- **Master of Arts (MA)** and **general "Master"** degrees follow
- **Professional degrees (MBA, MEd, etc.)** represent a smaller but significant segment
- **Long-tail degree** types exist but are less prevalent

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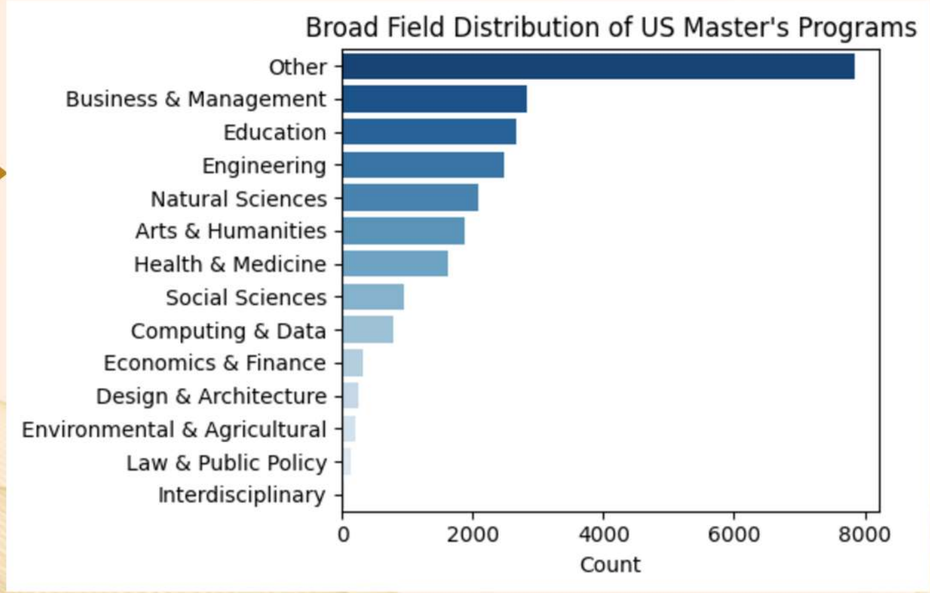
Exploratory Data Analysis (EDA)

Master's programs

Key Insights:

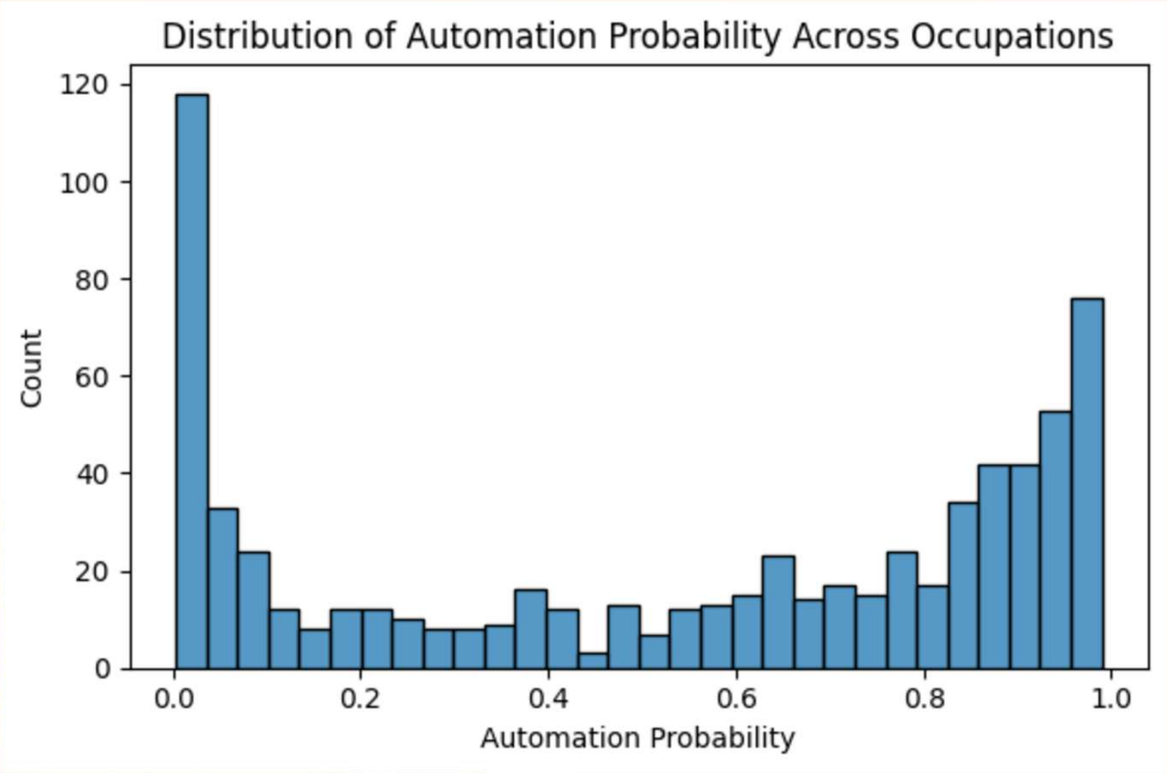
- Individual programs such as **Computer Science** and **Business Administration** are highly prevalent
- The long tail of specialized programs suggests **strong academic diversity**
- When grouped into broader fields, **Business & Management** emerges as the largest category
- **Engineering**, **Natural Sciences**, and **Health-related** disciplines also represent significant segments



Exploratory Data Analysis (EDA)

Automation

Minimum automation probability: 0.0028
Maximum automation probability: 0.9900
Mean probability: 0.5356



Key Observations:

- Automation probability **varies widely** across occupations
- A cluster of occupations shows very **low automation exposure**
- A significant portion exhibits **high automation susceptibility**
- Risk distribution is **highly uneven rather than uniform**

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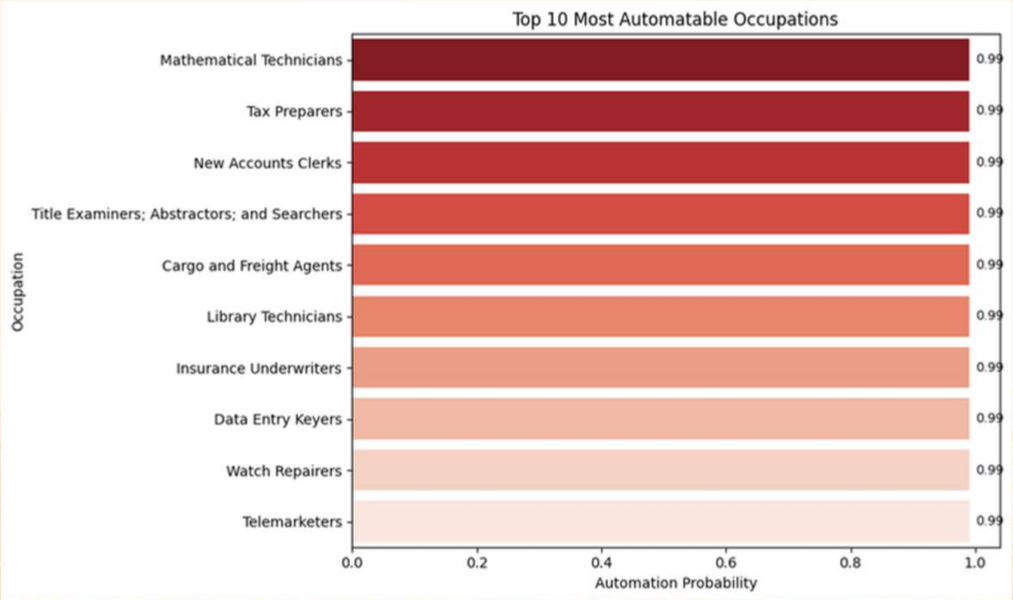
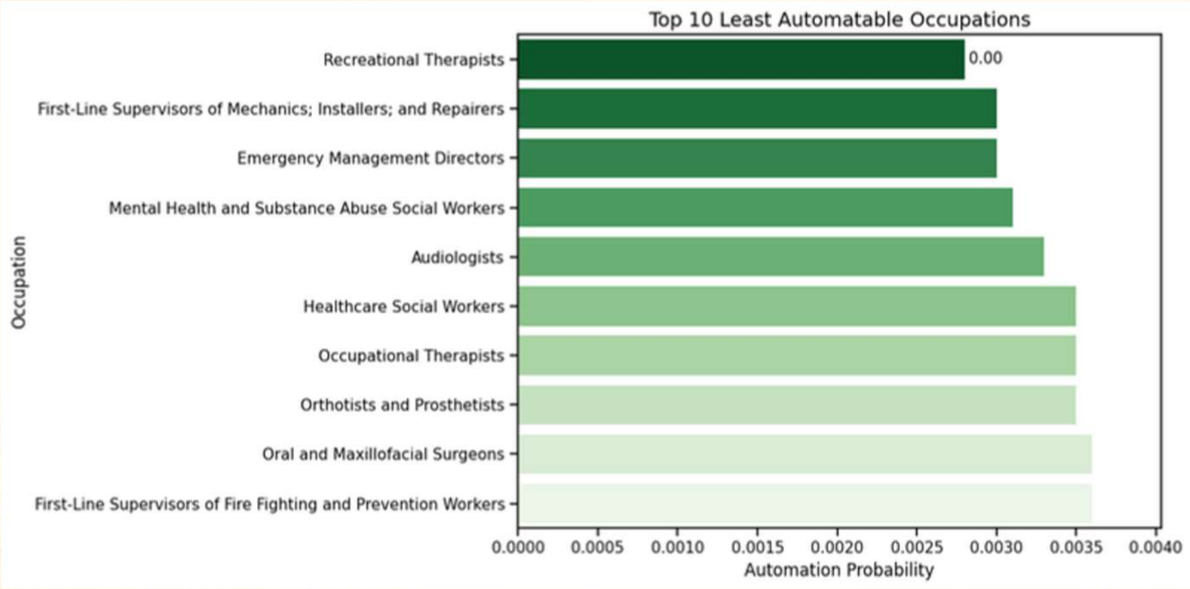
Exploratory Data Analysis (EDA)

Automation

Automation Probability Range: 0.00 – 0.99





Key Insights:

- Certain clerical and routine occupations show extremely high automation probability
- **Technical**, **healthcare**, and **supervisory roles** exhibit significantly **lower** automation exposure
- The contrast highlights structural differences in occupational vulnerability
- Automation risk is **occupation-specific** rather than **industry-wide**

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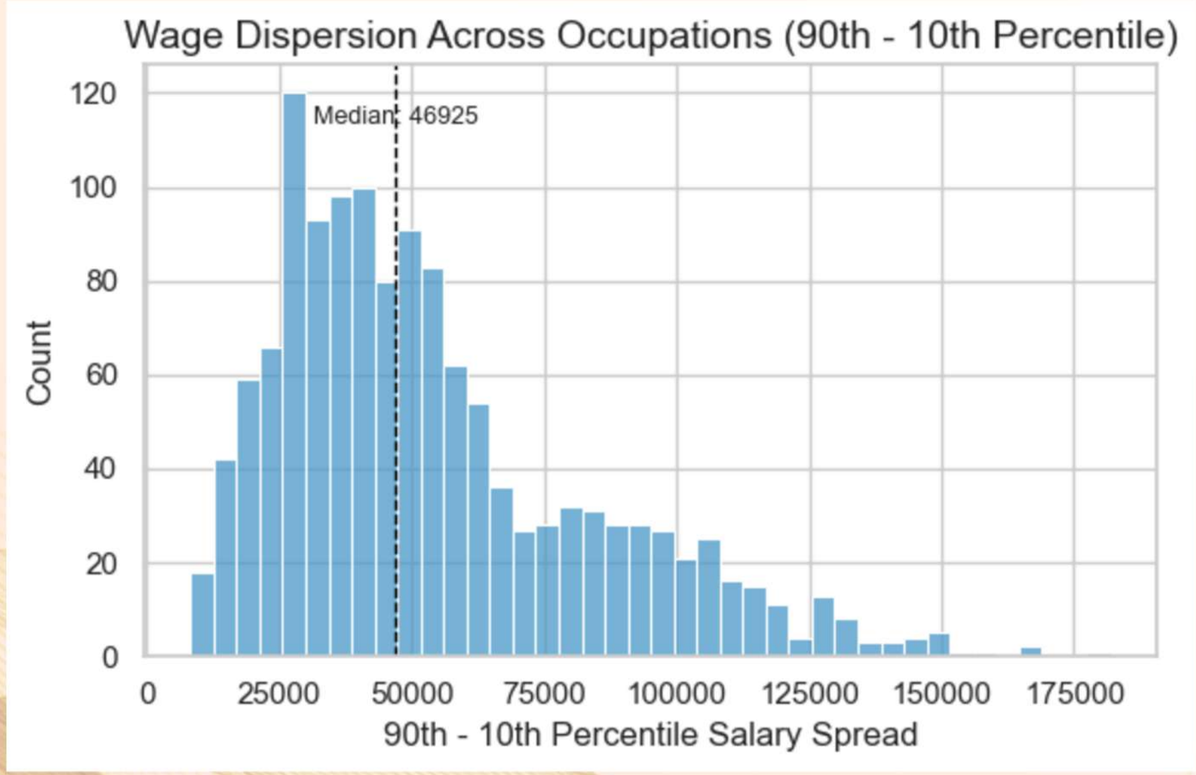
Exploratory Data Analysis (EDA)

Salary

Key Observations:

- Wage dispersion varies significantly across occupations
- **Median salary spread ≈ $46,925 (90th – 10th percentile)**
- Some occupations show tight wage bands (**stable income range**)
- Others exhibit large dispersion, indicating **higher income variability**

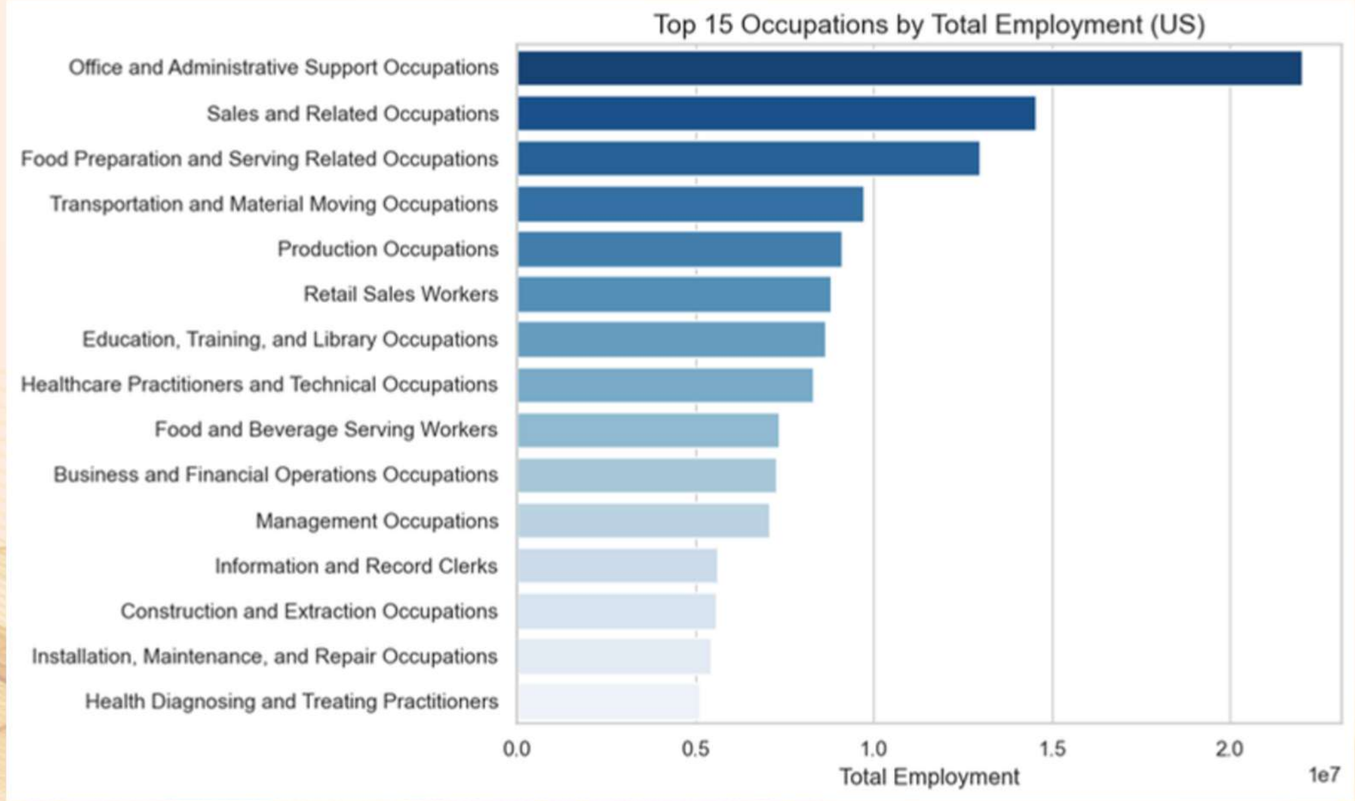


Exploratory Data Analysis (EDA)

Salary

Key Observations:

- Employment size varies substantially across occupation groups
- **Office & Administrative Support** is the **largest** employment category
- **Service**, **sales**, and **food-related occupations** dominate total workforce
- High employment volume indicates structural labor market importance



Graduate Career Stability Index (GCSI)

Definition: A composite index (0–100 scale) evaluating occupational career stability based on economic strength, labor market depth, and automation exposure.

Income Strength (35%) : Median annual salary of occupation

Employment Scale (25%): Total employment size (market depth)

Wage Stability (20%): Salary dispersion (P90 – P10 spread)

Automation Safety (20%) : 1 – Automation Probability

GCSI = (0.35×Income Strength) + (0.25×Employment Scale) + (0.20×Wage Stability) + (0.20×Automation Safety)

Analytical Techniques

1. **Descriptive**: Summarizing degree categories and job frequencies
2. **Diagnostic**: Explaining why certain paths are higher risk
3. **Prescriptive**: Recommending stable paths.

Methodological Framework

- **Data Cleaning & Harmonization**
- **Feature Engineering** (Salary Spread, Automation Risk, Employment Scale)
- **Normalization** (Min–Max scaling)
- **Ranking & Dashboard Visualization**
- **Weighted Index Construction**

Conclusion

This project aims to **reduce the guesswork** in choosing a Master's degree as jobs and skills continue to change rapidly. It uses data to **compare career paths**, highlight **which options are more stable** in the coming years, and identify the key skills that support **long-term employability**.

Thank You!
Any Questions?